🚀 Day 2/30: Python Fundamentals for AI/ML Beginners

Goal of Day 2:

Build a strong Python foundation before starting Data Science and Machine Learning.

Time Required: 3–4 Hours

📌 1. Variables and Data Types

Variables store data in memory.

Integer (int)

age = 20

print(age)

```
age = 20 print(age)
```

Output:

```
20
```

Float (decimal numbers)

cgpa = 8.5
print(cgpa)

```
cgpa = 8.5 print(cgpa)
```

Output:

```
8.5
```

String (text)

name = "Utkarsh"
print(name)

```
name = "Utkarsh" print(name)
```

Output:

```
Utkarsh
```

Boolean (True/False)

is_student = True
print(is_student)

```
is_student = True print(is_student)
```

Output:

```
True
```

2. User Input

Taking input from users:

name = input("Enter your name: ")

print("Hello,", name)

```
name = input("Enter your name: ")
print("Hello,", name)
```

Example:

```
Enter your name: Utkarsh Hello,
Utkarsh
```

📌 3. Operators

Arithmetic Operators

a = 10

b = 5

print(a + b)

print(a - b)

print(a * b)

print(a / b)

```
a = 10 b = 5 print(a + b) print(a - b)
print(a * b) print(a / b)
```

Output:

```
15 5 50 2.0
```

Comparison Operators

print(10 > 5)

print(10 == 5)

```
print(10 > 5) print(10 == 5)
```

Output:

```
True False
```

📌 4. Conditional Statements
If Statement
age = 18

if age >= 18:
print("You can vote")

If-Else
age = 16

if age >= 18:
print("Adult")

else:
print("Minor")
Output:
Minor

📌 5. Loops

For Loop

```
for i in range(5): print(i)
```

Output:

```
0 1 2 3 4
```

While Loop
count = 1

while count <= 5:
print(count)

count += 1

📌 6. Lists

Lists store multiple items.
skills = ["Python", "Git", "NumPy"]

print(skills)

```
skills = ["Python", "Git", "NumPy"]
print(skills)
```

Output:

```
['Python', 'Git', 'NumPy']
```

Adding Elements
skills.append("Pandas")

print(skills)

```
skills.append("Pandas") print(skills)
```

Output:

```
['Python', 'Git', 'NumPy', 'Pandas']
```

Sorting

numbers = [5, 2, 8, 1]

numbers.sort()

print(numbers)

```
numbers = [5, 2, 8, 1] numbers.sort()
print(numbers)
```

Output:

```
[1, 2, 5, 8]
```

List Slicing

numbers = [10, 20, 30, 40, 50]

print(numbers[1:4])

```
numbers = [10, 20, 30, 40, 50]
print(numbers[1:4])
```

```
[20, 30, 40]
```

📌 7. Dictionaries

Dictionaries store data in key-value pairs.

student = {
"name": "Utkarsh",
"branch": "AI/ML",
"year": 2
}

```
student = { "name": "Utkarsh", "branch": "AI/ML", "year": 2 }
```

Accessing Values

```
print(student["name"])
```

Output:

```
Utkarsh
```

Using get()

```
print(student.get("branch"))
```

Output:

```
AI/ML
```

Getting Keys

```
print(student.keys())
```

Output:

```
dict_keys(['name', 'branch', 'year'])
```

Getting Values

```
print(student.values())
```

Output:

```
dict_values(['Utkarsh', 'AI/ML', 2])
```

📌 8. Functions

Functions help reuse code.

Basic Function

def greet():

print("Hello AI Student!")

greet()

```
def greet(): print("Hello AI Student!")
greet()
```

Output:

```
Hello AI Student!
```

Function with Parameters
def greet(name):
print("Hello", name)

greet("Utkarsh")

```
def greet(name): print("Hello", name)
greet("Utkarsh")
```

Output:

```
Hello Utkarsh
```

Function with Return Value
def add(a, b):
return a + b

result = add(10, 20)

print(result)

```
def add(a, b): return a + b result = add(10, 20) print(result)
```

Output:

```
30
```

💻 Mini Project 1: Student Grade Calculator
marks = int(input("Enter marks: "))

if marks >= 90:
print("Grade A")

elif marks >= 75:
print("Grade B")

elif marks >= 50:
print("Grade C")

else:
print("Fail")

💻 Mini Project 1: Student Grade

Calculator

```
marks = int(input("Enter marks: ")) if
marks >= 90: print("Grade A") elif
marks >= 75: print("Grade B") elif
marks >= 50: print("Grade C") else:
print("Fail")
```

💻 Mini Project 2: Basic Calculator
a = int(input("First Number: "))
b = int(input("Second Number: "))

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)

💻 Mini Project 2: Basic Calculator

```
a = int(input("First Number: ")) b =
int(input("Second Number: "))
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
```

💻 Mini Project 3: To-Do List App

```
tasks = []

tasks.append("Learn Python")
tasks.append("Solve DSA")
tasks.append("Learn NumPy")

print("Today's Tasks:")

for task in tasks:
print(task)
```

💻 Mini Project 3: To-Do List App

```
tasks = [] tasks.append("Learn
Python") tasks.append("Solve DSA")
tasks.append("Learn NumPy")
print("Today's Tasks:") for task in
tasks: print(task)
```

Output:

```
Today's Tasks: Learn Python Solve
DSA Learn NumPy
```

🎯 Practice Problems

Solve these today:

Easy

1. Reverse a string
2. Find the maximum element in a list
3. Count vowels in a string
4. Check if a number is prime
5. Print a multiplication table

Medium

6. Check if a string is a palindrome
2. Find duplicate elements in a list
3. Create a simple login system using dictionaries

📚 Resources

YouTube

- Python Full Course for Beginners
- Python Practice Questions
- Python Mini Projects

Practice Platforms

- HackerRank (Python Basics)
- LeetCode Easy Problems
- CodeChef Beginner Problems

🎯 Day 2 Challenge

By the end of Day 2, you should be able to:

✅ Write basic Python programs

✅ Use lists and dictionaries confidently

✅ Create functions

✅ Build simple console projects

✅ Solve 5 Python problems independently

🚀 Tomorrow (Day 3)

Topic: Git & GitHub Basics

You'll learn:

- What is Git?
- What is GitHub?
- Installing Git
- Essential Commands
- Creating Your First Repository
- Pushing Code to GitHub

Comment "DAY2" on Instagram to get this complete roadmap for free!